IN1020 GRUPPETIME

GRUPPE 3 – JUSTAS BALTRUKONIS

BOOLSK ALGEBRA OG PORTER

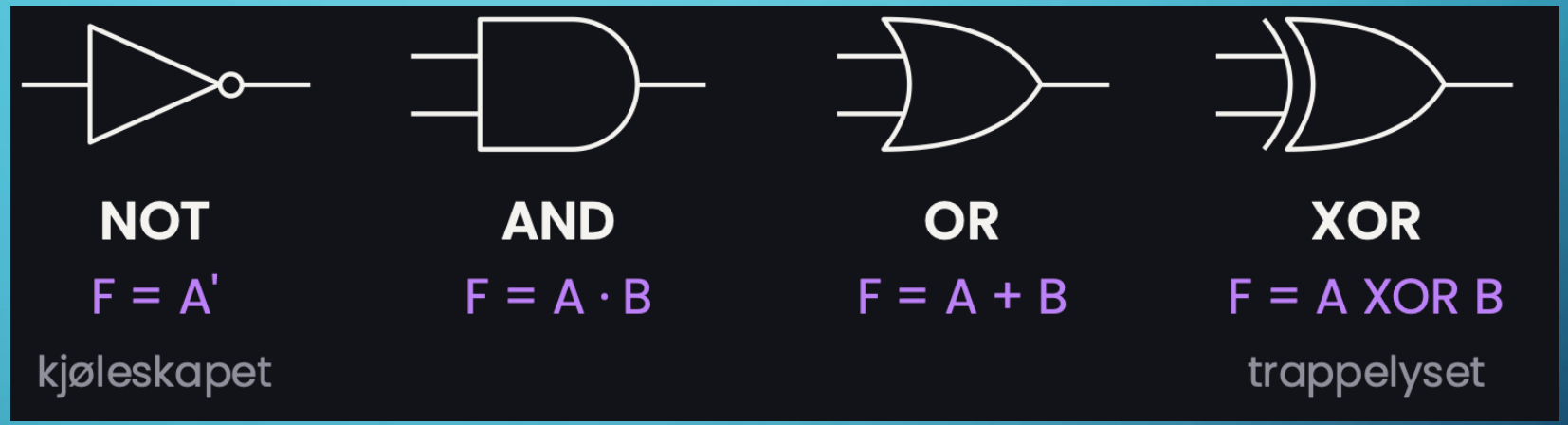
- De morgans
- Porter
- Verditabeller
- Transistor
- Halvadder

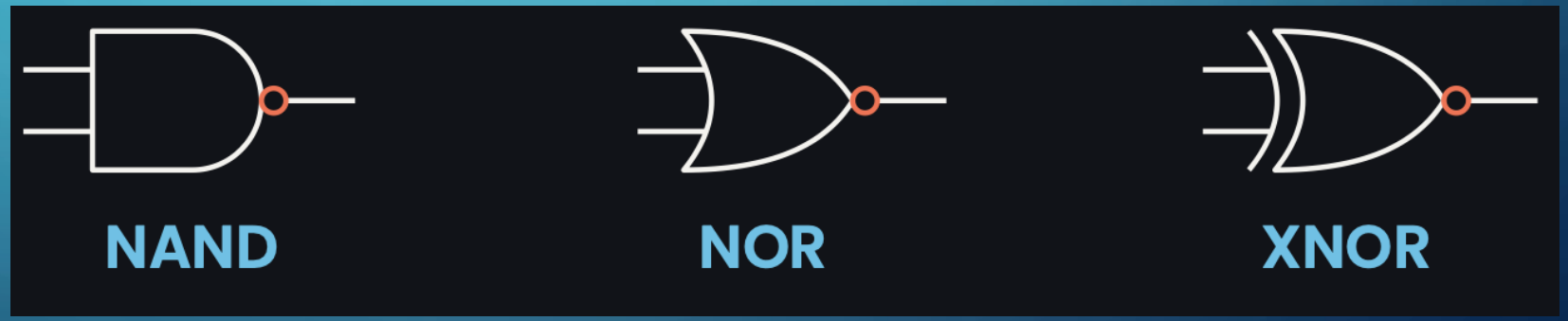
PORTER

PORTER

- AND
- OR
- NOT
- XOR

- NAND
- NOR
- XNOR



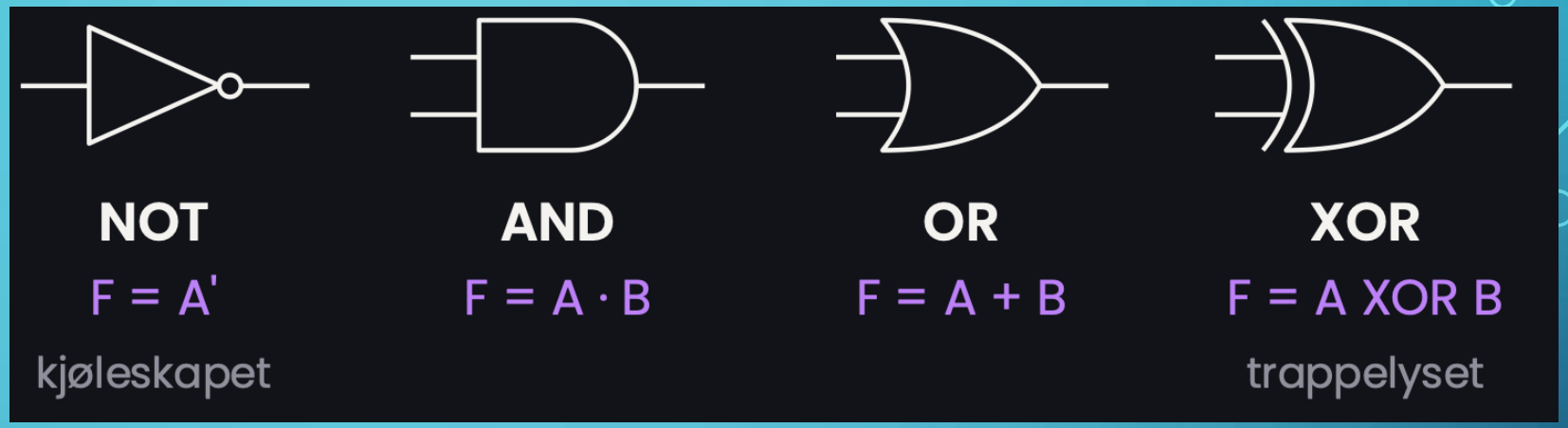


VERDITABELL

- For F så putter man inn formelen
- A og B representerer bit verdier med 2 bits

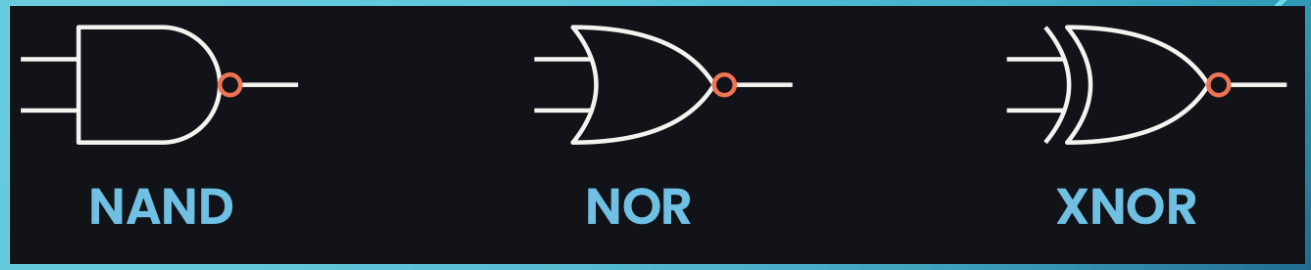
A	B	F
0	0	-
0	1	-
1	0	-
1	1	-

PORTER



- AND – begge
- OR – minst en eller begge
- NOT – invers/ motsatt (inverter
- XOR – eksklusiv OR – en, men ikke begge
  - Huskeregel: 1 (true) hvis inngangene er ulike.
  - Trappelyset – endrer tilstand uansett hvordan den andre står
  - Binær addisjon – 1+1 gir 0 med 1 i mente

PORTER – INVERTERTE



- Hvordan?
  - Utfør den originale porten og flip svaret!

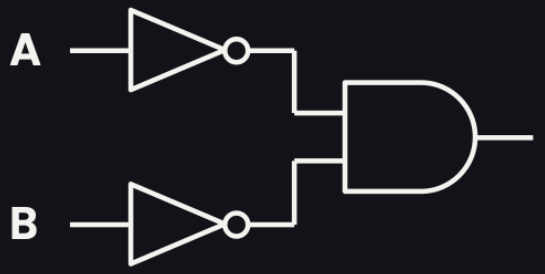
- NAND – invertert AND eller NOT AND
- NOR – NOT OR
- XNOR – NOT XOR (exclusive or) – sjekker likhet

FORMLER

- Utifra formelen er verditabellen enkel

PORT	FORMEL
NOT	$F = A'$ - inverter
AND	$F = A \cdot B$
OR	$F = A + B$
XOR	$F = A \oplus B$
NAND	$F = (A \cdot B)'$
NOR	$F = (A + B)'$
XNOR	$F = (A \oplus B)'$

HVOR INVERTERES DET

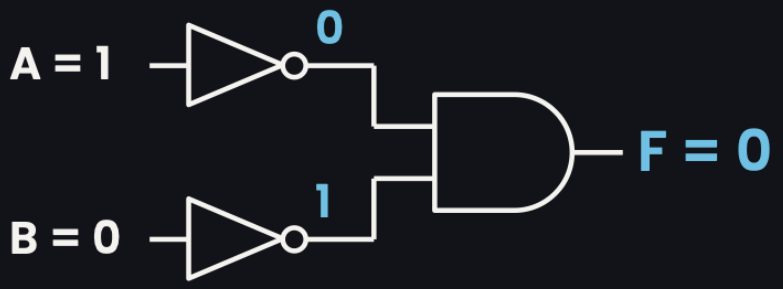


tre porter

A	B	A'	B'	F
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0



én port: NOR



$A' \cdot B'$

hver inngang snus for seg



$(A \cdot B)'$

hele svaret snus

DE MORGANS LOV

- Boolsk algebra sin pytagoras

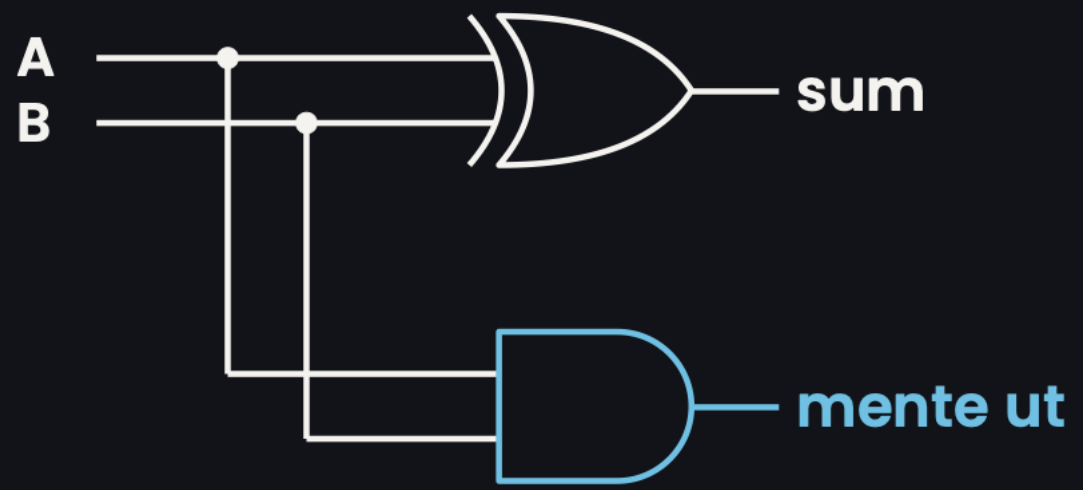
- NAND = INV a OR INV b
- NOR = INV a AND INV b

$$(A \cdot B)' \equiv A' + B'$$

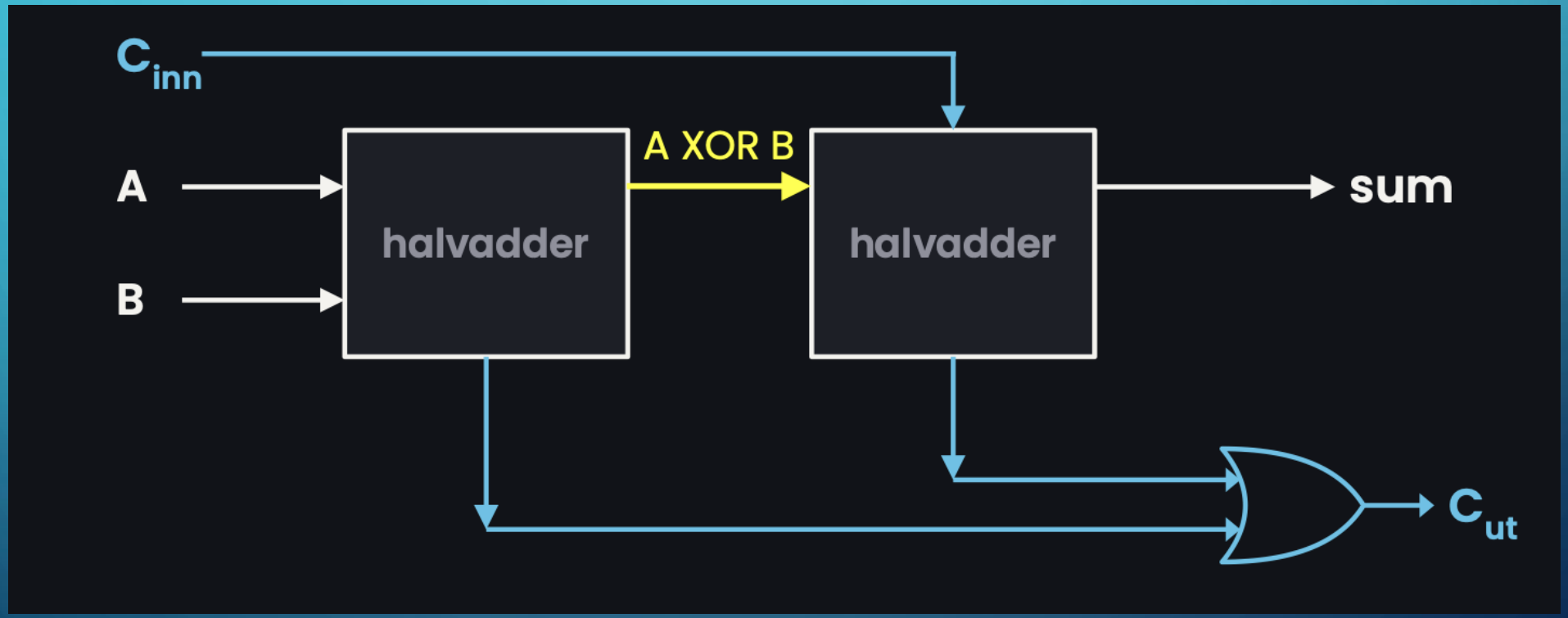
$$(A + B)' \equiv A' \cdot B'$$

HALVADDER

A	B	sum	mente
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



FULLADDER



TRANSISTOR

- Veldig enkelt – enten så slipper den strøm gjennom eller ikke
  - 0 eller 1
  - Av eller på
- Alt som skjer i maskinen er strøm – fysiske signaler
- Transistoren bestemmer om strømmen går eller ikke

FERDIG

- Tegn masse tabeller så det sitter
- Universet